Fire & Smoke Vision Core: A Multi-Module Deep Learning Approach to Fire Detection, Localization & Severity Assessment

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Abstract

A multi-module computer-vision system that detects fire and smoke in images. It unifies five deep-learning tasks — classification (ResNet18), detection and segmentation (YOLOv8), Grad-CAM explainability and a severity score — trained from one dataset of 5,000 images. On unseen test data it reaches 97.5% accuracy, 0.79 detection mAP and 0.79 mask mAP.

Introduction

Fire and smoke threaten life and property; early visual detection is vital for surveillance and warning systems. Most solutions only classify 'fire or not' and cannot show where the fire is or how severe it is.

Objectives:

- Add localization, segmentation and risk scoring.
- Train all tasks from one dataset.
- Explain predictions with Grad-CAM.
- Provide an easy web interface.

Methodology

Dataset
5,000 images · LabelMe polygons
→ classification + detection + segmentation labels

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Input Image
JPG / PNG

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Preprocessing
Resize · Normalize · Augment

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Module 1 — Classification
ResNet18 → fire / normal

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Module 2 — Detection
YOLOv8 → bounding boxes

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Module 3 — Segmentation
YOLOv8-seg → pixel masks

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Module 4 — Explainability
Grad-CAM heatmap

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Module 5 — Severity
% frame on fire → risk level

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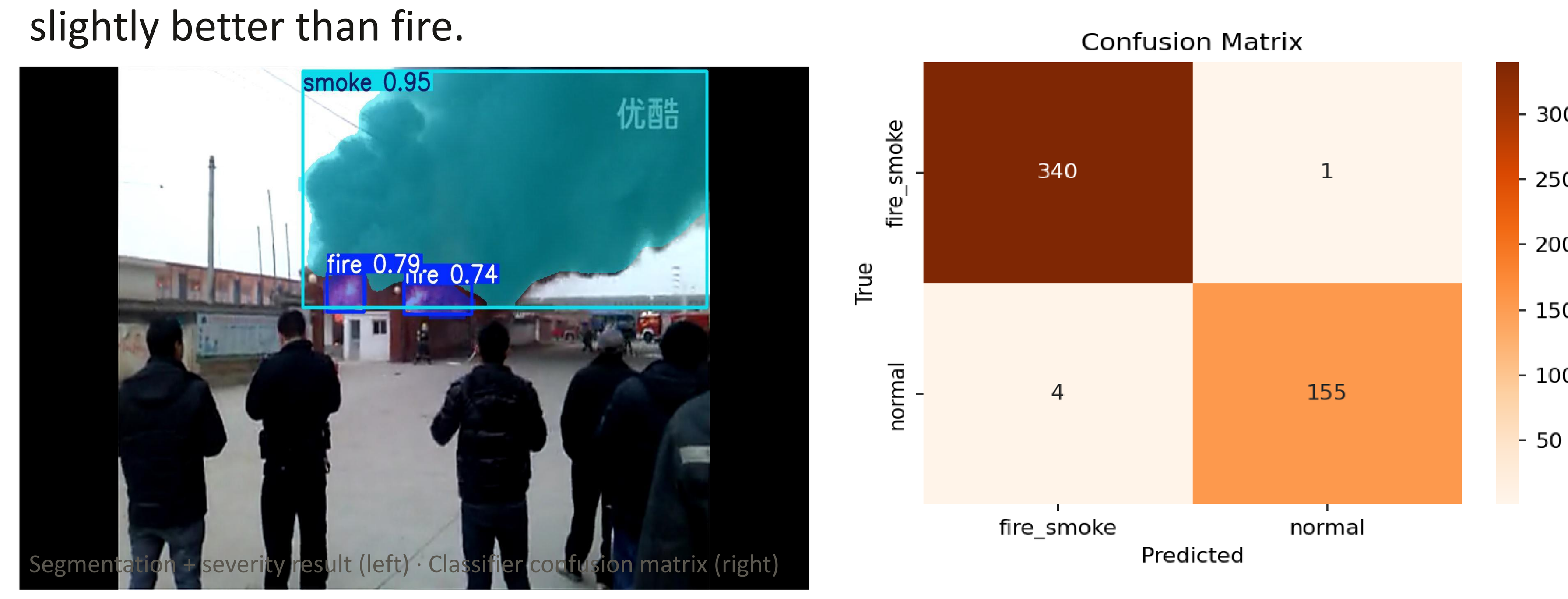
React Web Application
FastAPI backend serves all modules

Related Work

Study	Focus	Strength	Limitation
Muhammad+ 2018	CNN fire classification	High accuracy	No localization
Li & Zhao 2020	Faster R-CNN fire	Bounding boxes	Heavy; no masks
Jadon+ 2019 (FireNet)	Lightweight CNN	Real-time	Classify only
Shamsoshoara 2021 (FLAME)	Aerial fire seg.	Pixel masks	Single task
YOLOv8 (Ultralytics) 2023	General det./seg.	Fast, SOTA	Not fire-specific
This work, 2025	5 unified modules + XAI	Explainable, end-to-end	Single-domain data

Results

Evaluated on a held-out test set of 500 unseen images. Test scores match or beat validation, so the models generalize without overfitting. Smoke localizes slightly better than fire.



Segmentation + severity result (left) · Classifier confusion matrix (right)

Module	Metric	Score
Classification (ResNet18)	Test accuracy	97.5%
Detection (YOLOv8)	mAP@50	0.787
Segmentation (YOLOv8-seg)	Mask mAP@50	0.792
Detection per-class	fire / smoke mAP@50	0.75 / 0.80
Severity	Frame coverage	NONE→CRITICAL

Conclusion & Future Directions

One dataset powers five fire-analysis tasks in a single explainable web tool, reaching 97.5% accuracy and ~0.79 mAP. Future work: higher recall, real-time video, and edge deployment.

References

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